Week 1.3 — TriggerFish Build Begins — Working vs. Crafted

Sunday, June 28, 2026 · 3:00 – 5:15 PM

Lesson 1.3 was the start of the actual TriggerFish build. The SpaceWings practice kit from 1.2 is behind us; the real ROV control board sat on the granite counter today, out of the kit, waiting. Eight kids showed up — Maddox, Frankie, Rebecca, Isaac, Liam, Luke, Tyler, and Trenton (Trenton stepped out at 3:45). Meeting ran 3:00 to 5:15. The day's work: kit opened, ten to fifteen components soldered onto the board per the SeaMATE TriggerFish ROV Guide (the two 1000µF / 16V capacitors with polarity verified, the fuse holder, the 3 A fuse, and the next block from the slide), ribbon wires soldered to the joysticks, and the 150A high-precision watt meter and power analyzer added to the bench with extended leads. One process note worth keeping: the existing iron tip was oxidized to the point that the standard 650°F wasn't flowing solder cleanly. We swapped to a slightly wider tip and bumped the temperature to 720°F to compensate. The lesson of the day wasn't a technique; it was a frame. The team distinguished a 'working' build from a 'crafted' build, and named the connection between craftsmanship and reliability. Officer elections stay deferred until the summer TriggerFish build is wrapped.

8	10–15	650 → 720°F	150 A
STUDENTS	COMPONENTS SOLDERED	IRON TEMP (NEW TIP)	WATT METER ON BENCH

WHAT WE LEARNED

- Working and crafted both pass a multimeter check. Only crafted survives vibration, heat, water ingress, and the trip to the pool deck. Reliability is a function of craftsmanship, not luck.
- Craftsmanship has a specific checklist on this board: polarity verified before commit, joints clean and shiny, components seated flush, leads clipped flush, heatshrink with strain relief.
- If you're not sure a joint is crafted, ask whether you'd ship it to the regional and bet a season on it. If the answer is "probably," redo it.
- Iron tips don't last forever. When solder won't flow at the standard temp, the tip is the variable — not the operator. Change it, document, bump the setpoint. Tonight: 650 → 720°F.

STUDENT LEARNING OUTCOMES

- A working definition of "craftsmanship" that isn't vague effort — polarity check, clean joints, flush seating, clipped leads, heatshrink with strain relief.
- First time handling live ROV-grade components — the control board, the 1000µF capacitors, the watt meter that reads real current.
- The discipline of working against a printed-and-projected reference (the SeaMATE Guide). Build steps come from the page, not from memory.
- Equipment care is part of the engineering. Tonight the iron tip degraded mid-build; the response was to notice, document, change, adjust — not push through and hope.

LOOKING AHEAD

Class 1.4 continues the TriggerFish build — next blocks of components on the control board, working toward a powered-up smoke test later in Phase 1. SeaMATE Guide stays open on the iPad.

DECISIONS MADE

- Officer elections deferred until the summer 1.01 TriggerFish build is complete. Roles get assigned based on what the team actually does well during the build.
- Polarity gets verified before any electrolytic cap touches the board. Tuition for getting it wrong on a 1000µF cap is high.
- The SeaMATE TriggerFish ROV Guide is the team's procedural reference of record for Phase 1. Every step verified against the slide before any iron touches the board.
- When the iron tip oxidizes past clean-flow at 650°F, swap the tip and bump the temp until the new tip flows cleanly. Tonight: wider tip, 720°F.

ACTION ITEMS

Continue the TriggerFish control-board build at Lesson 1.4
Sunday cohort · 7/5

Nominate a photographer for the rest of the summer build
Advisor (until CEO is elected) · by 1.4

Start a parts-inventory tracker for the TriggerFish kit
Advisor (until CFO is elected) · by 1.4

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